

UAV Audio Detection and Identification Using Short-Time Fourier Transform Spectrograms with Deep Learning Models

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Abstract—Unmanned Aerial Vehicles (UAVs), commonly known as drones, offer immense potential across diverse sectors but also raise significant security concerns due to their accessibility and potential misuse. Effective detection and identification of drones are therefore essential to mitigate associated risks. This study investigates the use of Short-Time Fourier Transform (STFT)-based audio spectrograms combined with deep learning models for drone detection and classification based on acoustic signatures. We evaluate the performance of Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Convolutional Recurrent Neural Networks (CRNN) in a multi-class classification setting. Our findings show that STFT spectrograms, when paired with deep learning architectures, yield high accuracy in distinguishing between drone models and non-drone sounds. Among the models tested, CNN consistently demonstrates the best trade-off between performance and computational efficiency. This work contributes a comparative analysis of deep learning models using acoustic features and highlights their potential for reliable UAV identification, with future application toward real-time and scalable security systems.

Index Terms—Drone, UAV, drone audio, drone detection, drone identification, CNN, RNN, CRNN, STFT, spectrogram, deep learning, machine learning, data augmentation, entropy thresholding, artificial intelligence.

I. INTRODUCTION

Drones have rapidly evolved from military tools into versatile assets across numerous industries and civilian applications. They are now widely used for last-mile delivery, precision agriculture, infrastructure surveying, disaster response, and aerial cinematography [6]. This widespread adoption has contributed to rapid market growth, with forecasts predicting the global drone industry will reach approximately 13 billion USD by 2025, representing a 200% increase within five years [7].

However, the growing ubiquity of drones also brings increased risk. While some countries enforce strict regulations or bans on drone sales and operations, others permit easy acquisition with limited oversight [8]. This accessibility raises serious concerns about misuse, such as unauthorized surveillance, smuggling, and encroachment on restricted airspaces. As these incidents continue to rise, the need for robust and scalable drone detection and identification systems becomes increasingly urgent.

Traditional detection systems—such as optical sensors, radar, and radio frequency (RF) analyzers—face significant limitations, especially in low-light or urban environments and when targeting small, fast-moving UAVs. To address these limitations, researchers have increasingly explored audio-based detection methods, capitalizing on the fact that drone models emit distinct acoustic signatures during operation [2], [4]. These sound patterns, captured as time-frequency features, offer a promising modality for drone classification.

One effective way to visualize these acoustic signatures is through *Short-Time Fourier Transform (STFT)* spectrograms, which represent how frequencies evolve over time. While STFTs are sensitive to noise and may lose fine-grained temporal resolution, they provide a robust starting point for pattern recognition when paired with deep learning (DL) models. In this work, we explore the use of STFT-generated spectrograms to classify UAVs in noisy environments.

Specifically, we analyze the performance of three deep learning architectures—*Convolutional Neural Networks (CNNs)*, *Recurrent Neural Networks (RNNs)*, and *Convolutional Recurrent Neural Networks (CRNNs)*—within a multi-class classification framework. By training these models on spectrogram features extracted from both drone and non-

drone audio, we assess their ability to distinguish UAV types under real-world conditions.

Our results suggest that STFT-based spectrograms are highly effective for drone identification, with CNNs offering a strong balance between accuracy and computational efficiency. This research contributes a comparative evaluation of deep learning strategies for audio-based UAV classification and lays the foundation for real-time, scalable solutions in security and surveillance applications.

II. RELATED WORK

Various approaches have been explored for drone detection, including radar, LiDAR, computer vision, and audio-based techniques. Radar-based systems perform well for large aerial objects such as aircraft or missiles [1], [9]; however, they are often ineffective at detecting small, agile UAVs. LiDAR-based methods also suffer from limitations, particularly under adverse weather conditions such as rain or fog [10].

Computer vision techniques, often enhanced by machine learning, have been widely used for drone tracking. Algorithms such as Support Vector Machines (SVM), Channel and Spatial Reliability Tracking (CSRT), and Kernelized Correlation Filters (KCF) have demonstrated strong performance in object recognition tasks [21], [22]. Further advancements, such as parallel Convolutional Neural Networks (CNNs), have improved detection accuracy in object classification [23]. However, these methods are still affected by poor lighting, occlusions, and environmental variability [11].

To address these limitations, researchers have increasingly investigated drone detection using audio signals. Audio-based detection offers several benefits: it is less reliant on line-of-sight conditions, often more cost-effective than radar or vision systems, and suitable for diverse environmental settings [1], [2]. Early approaches analyzed low-level audio features such as Short-Time Energy, Temporal and Spectral Centroid, Spectral Roll-off, and Mel-Frequency Cepstral Coefficients (MFCCs) [2]. Techniques like Linear Predictive Coding (LPC) combined with Zero Crossing Rate and Frequency Slope have also shown effectiveness in detecting large UAVs [15].

With the rise of machine learning, these features have been used to train models for more robust classification. Prior work has applied MFCCs and Short-Time Fourier Transform (STFT)-based preprocessing alongside models such as K-Nearest Neighbors and classical classifiers [4], [17]. These efforts have evolved into leveraging 2D spectrograms as visual inputs to train deep learning models [13]. Notably, Al-Emadi et al. [13] explored CNN, RNN, and CRNN architectures for drone audio detection using deep learning, presenting a foundational study in this area.

A comparison of these diverse drone detection approaches, including their advantages and limitations, is summarized in Table I. This overview highlights the trade-offs between traditional and modern techniques, illustrating the growing

effectiveness of audio-based deep learning methods in complex environments.

TABLE I
COMPARISON OF DRONE DETECTION TECHNIQUES IN PRIOR WORK

Method	Advantages	Limitations
Radar-based [1], [9]	Long-range detection; effective for large objects	Poor with small or low-flying drones
LiDAR-based [10]	Accurate 3D spatial mapping	Sensitive to weather; expensive
Vision + ML (SVM, CSRT, KCF) [21], [22]	Good performance in clear conditions	Affected by lighting, fog, occlusion
Parallel CNN [23]	High accuracy in object detection	Resource-intensive; not weather-resilient
Audio Features (MFCC, LPC) [2], [15]	Effective in low-light or occluded settings	Affected by ambient noise
Spectrogram + classical ML [4], [17]	Enables visual analysis of sound	Limited learning capacity
Spectrogram + DL (CNN, RNN, CRNN) [13]	Captures spatial-temporal features well	Requires large diverse datasets

Building upon these efforts, our work focuses on enhancing UAV audio detection through improved dataset diversity and a comparative evaluation of deep learning models. We expand the dataset using audio from various drone types, including real-world and online sources. Using STFT spectrograms as inputs, we analyze the effectiveness of CNN, RNN, and CRNN models within a unified framework. Unlike previous work that evaluated a single architecture, we provide a comparative analysis across multiple deep learning models to assess their respective strengths and performance under varying conditions.

III. PROPOSED METHODOLOGY

Our proposed framework for drone audio detection is structured into five core stages: (1) data acquisition, (2) dataset formulation, (3) signal preprocessing via Short-Time Fourier Transform (STFT), (4) deep learning model design, and (5) model training and evaluation. The entire pipeline is designed to explore and compare the performance of CNN, RNN, and CRNN architectures for audio classification using spectrogram representations.

A. Data Acquisition

To ensure a diverse and realistic dataset capable of generalizing across drone models and environmental conditions, we sourced audio data from both real-world recordings and online repositories:

- **Experimentally Captured Audio:** Drone sounds were captured in controlled settings using an iPhone 13's built-in microphone. Devices included the CrazyFlie and

DJI Phantom 4 Pro. This provided high-quality, clean audio useful for initial training.

- **YouTube Extracted Audio:** Audio streams from publicly available drone flight videos were extracted using FFMPEG, targeting DJI Mavic Pro and Matrice 100 drones. These clips contain realistic noise profiles such as wind and distant speech.
- **Online Public Datasets:** We included pre-labeled drone audio from datasets such as those found on Kaggle and research repositories, specifically for drones like Parrot Bebop and Mambo.
- **Non-Drone Environmental Sounds:** To minimize false positives and increase model robustness in real-world deployment, we collected background audio including bird chirps, rainfall, engine noise, street traffic, and human voices.
- **Data Augmentation:** To address class imbalance, pitch shifting was applied using Librosa's 'pitch shift()' function. For example:

- CrazyFlie: ± 1 semitone
- Matrice 100, Mavic, Phantom: ± 2 and ± 1 semitones

This increased variation in tonal characteristics while preserving temporal structure, thus helping models generalize better to similar but unseen audio.

B. Audio Dataset Description

The final dataset comprises one-second mono audio clips labeled into seven classes (six drone types and one non-drone category). Each clip was sampled at 16,000 Hz and normalized to ensure amplitude consistency.

Table II presents the sample distribution:

TABLE II
NUMBER OF AUDIO SAMPLES PER CLASS

Audio Class	Train Samples	Test Samples
Bebop	420	104
Crazyflie	210	52
Mambo	420	104
Matrice100	200	50
Mavic	200	50
Non-Drone	1434	357
Phantom	204	51

To enhance robustness, drone clips include both clean and noisy recordings. Non-drone audio was selected to challenge the models in realistic auditory scenes where drone signals may be masked or distorted.

C. Data Preprocessing

We employed the Short-Time Fourier Transform (STFT) to transform each 1-second audio clip into a time-frequency domain representation. STFT captures transient and harmonic components by computing the Fourier Transform on overlapping windows:

$$X(m, k) = \sum_{n=0}^{N-1} x(n + mH) \cdot w(n) \cdot e^{-i2\pi nk/N}$$

where:

- $x(n + mH)$ is the windowed signal,
- $w(n)$ is the Hann window,
- $N = 2048$ is the FFT window size,
- $H = 256$ is the hop size.

An example STFT spectrogram of Bebop drone audio is shown in Fig. 1.

This setting yields a good trade-off between frequency resolution (7.8 Hz) and time resolution (16 ms), ideal for detecting drone audio features such as rotor blade harmonics and motor whining.

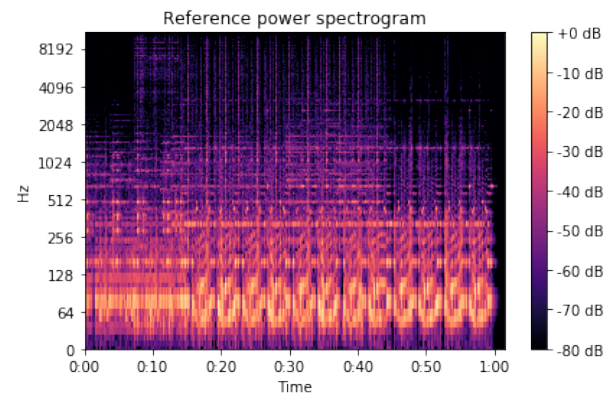


Fig. 1. The spectrogram illustrates the frequency content over time, highlighting the unique acoustic signature of the drone.

Librosa's 'stft()' function was used with:

- Sampling rate: 16,000 Hz
- FFT window size: 2048
- Hop length: 256

The resulting spectrograms had a shape of approximately 128×55 , representing 128 frequency bins over 55 time slices. These were treated as grayscale images and normalized before being passed into deep learning models.

D. Model Architectures and Evaluation Metrics

To explore the strengths of different neural paradigms, we implemented and compared three models:

1) **CNN-STFT Architecture:** This model is designed to exploit spatial locality and visual patterns in spectrograms:

- 3 convolutional layers with increasing filters (32, 64, 128), each with 3×3 kernels and ReLU activation
- MaxPooling (2×2) layers after each convolution
- Flatten layer
- Dense (128, ReLU) + Dropout (0.5)
- Dense (7, Softmax) for classification

2) **RNN-STFT Architecture:** This model captures temporal dependencies in the spectrogram columns:

- Input reshaped to time-step sequences
- 2 Bidirectional RNN layers (32 units each, ReLU)
- Batch normalization to stabilize training
- Dense (32, ReLU) + Dropout (0.3)
- Dense (7, Softmax)

3) **CRNN-STFT Architecture:** Combining spatial and temporal modeling:

- CNN feature extractor (same as CNN-STFT up to flatten)
- Output reshaped and passed into Bidirectional RNN (32 units)
- Batch normalization
- Dense layers same as RNN-STFT

All models were trained using:

- Batch size: 32
- Optimizer: Adam
- Loss: Categorical Crossentropy
- Epochs: 185

We evaluated the models on four metrics: Accuracy, Precision, Recall, and F1-Score, calculated per class and averaged.

E. Experimental Framework and Model Complexity

Each 1-second clip produces a spectrogram with height 128 (from FFT size) and width calculated as:

Thus, each input to the model has shape $128 \times 56 \times 1$.

Table III summarizes the input and parameter count for each architecture:

TABLE III
MODEL ARCHITECTURES AND PARAMETER COUNT

Model	Input Shape	Parameters
CNN-STFT	$128 \times 56 \times 1$	3,723,479
RNN-STFT	128×56	402,711
CRNN-STFT	$128 \times 56 \times 1$	7,343,831

The CRNN model is the most complex due to the combined CNN and RNN layers, but it is expected to offer superior performance by learning both spectral patterns and temporal transitions.

IV. RESULTS

A. Overall Experimental Metrics

Table IV presents a comprehensive evaluation of the CNN, RNN, and CRNN models trained on STFT spectrogram inputs. Among the three architectures, the CNN-STFT model achieved the highest overall performance across all four metrics—Accuracy (96.48%), Precision (96.98%), Recall (96.35%), and F1-Score (97.02%). These results indicate that the CNN model not only identifies drone audio with high reliability but also maintains balanced performance in detecting true positives and minimizing false classifications.

The performance comparison in Table IV is visually summarized in Figure 2. The CNN model's superior generalization can be attributed to its ability to extract robust spatial

TABLE IV
OVERALL PERFORMANCE OF STFT-BASED MODELS

Model	Accuracy	Precision	Recall	F1-Score
CNN-STFT	96.48%	96.98%	96.35%	97.02%
RNN-STFT	93.62%	93.86%	93.62%	93.80%
CRNN-STFT	95.05%	95.17%	94.92%	94.88%

features from spectrogram inputs, outperforming both the RNN and CRNN models despite using fewer parameters than the CRNN model.

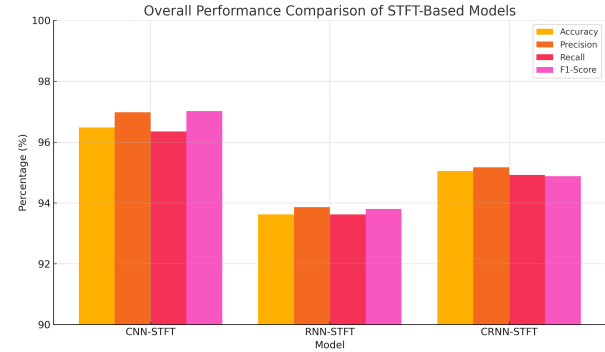


Fig. 2. Comparison of overall performance metrics (Accuracy, Precision, Recall, and F1-Score) for CNN, RNN, and CRNN models trained on STFT spectrograms. Each model was trained for 185 epochs with a batch size of 32 using spectrograms of size 128×56 .

From Figure 2, CNN-STFT outperformed CRNN-STFT by 1.4–2.1% across all metrics and achieved 2.9% higher accuracy than RNN-STFT. Despite its superior performance, CNN used 49.3% fewer parameters than CRNN (Table III), making it both accurate and computationally efficient.

B. Class-by-Class Analysis

To further investigate model behavior, we examined the class-wise performance of each architecture. Tables V, VI, and VII summarize accuracy, precision, recall, and F1-Score for each class in the test set.

Table V shows that CNN-STFT achieved perfect accuracy and precision for classes like *Crazyflie*, *Matrice100*, and *Mavic*, with near-perfect performance across all other drone classes and the *NonDrone* category. Notably, it achieved a 100% F1-score for *Crazyflie* and *Matrice100*, and maintained high precision even in mixed-sound scenarios (e.g., *Bebop* and *Mambo*).

TABLE V
CNN-STFT MODEL CLASS-WISE PERFORMANCE

Class	Accuracy	Precision	Recall	F1-Score
Bebop	99.04%	94%	99%	96%
Crazyflie	100%	100%	100%	100%
Mambo	84.62%	91%	85%	88%
Matrice100	100%	100%	100%	100%
Mavic	100%	100%	99%	99%
NonDrone	97.48%	97%	97%	97%
Phantom	98.04%	100%	98%	99%

Table VI presents RNN-STFT’s per-class metrics. Although it performed well on classes like *Crazyflie* and *Phantom* (100% accuracy and F1-score), it underperformed for the *Mambo* class, with a comparatively lower F1-score of 81%. This suggests that the RNN model may struggle to handle variations in background noise or spectral ambiguity.

TABLE VI
RNN-STFT MODEL CLASS-WISE PERFORMANCE

Class	Accuracy	Precision	Recall	F1-Score
Bebop	95.19%	91%	95%	93%
Crazyflie	100%	100%	100%	100%
Mambo	75.00%	88%	75%	81%
Matrice100	94.00%	98%	94%	96%
Mavic	100%	100%	100%	95%
NonDrone	95.80%	95.8%	96%	95%
Phantom	100.00%	100%	100%	96%

Table VII shows that CRNN-STFT performs better than RNN-STFT on average and reaches consistent results across most classes. It matched or exceeded CNN on some challenging classes such as *Mambo*, but fell slightly behind CNN-STFT overall.

TABLE VII
CRNN-STFT MODEL CLASS-WISE PERFORMANCE

Class	Accuracy	Precision	Recall	F1-Score
Bebop	97.12%	92%	97%	94%
Crazyflie	100%	98%	100%	99%
Mambo	87.50%	88%	88%	88%
Matrice100	90.00%	100%	90%	95%
Mavic	96.00%	100%	96%	98%
NonDrone	96.08%	97%	96%	97%
Phantom	98.04%	89%	98%	93%

In summary, the CNN-STFT model demonstrated the most consistent and highest performance across the majority of drone and non-drone classes, whereas RNN and CRNN occasionally struggled with variability in background audio.

C. Comparison with Prior Work

Table VIII benchmarks our models against prior studies by Al-Emadi et al. [13] and Kolamunna et al. [14]. Notably, while prior research was constrained to three classes, our model handles seven classes (six drone types and one non-drone), significantly increasing classification difficulty and real-world applicability.

TABLE VIII
COMPARISON WITH PRIOR WORK

Model	Our Accuracy	Prior Accuracy
CNN-STFT	96.48%	94.30% [13]
RNN-STFT	93.62%	57.16% [13]
CRNN-STFT	95.05%	92.65% [13]
Phantom Class	97–100%	68–70% [14]

As shown in Table VIII, our CNN-STFT model outperformed previous CNN-based implementations by over 2%. Most notably, the RNN-STFT model shows a dramatic

improvement of over 36% in accuracy compared to prior RNN approaches. Additionally, our detection accuracy for the Phantom class (up to 100%) is substantially higher than the 68–70% reported by Kolamunna et al., even under mixed-audio conditions.

These results validate the effectiveness of our dataset formulation, augmentation strategies, and model training pipeline for generalizable UAV audio detection in complex environments.

V. CONCLUSION

In this work, we propose an audio-based approach for UAV identification. To address our initial challenge of limited drone audio data, we constructed a comprehensive dataset that includes recordings from six different drone models. These audio files were obtained through experimental recordings, YouTube, online databases, and pitch-shifting techniques. This dataset also introduced the smaller and more maneuverable drone model *Crazyflie* for drone identification tasks.

Next, we trained three model architectures on STFT spectrogram datasets based on this audio dataset. The models demonstrated strong performance by achieving class accuracies of 90% or higher in six out of seven categories. Consequently, our models either maintained or exceeded the results reported in the existing literature, which shows both generalizability and effectiveness across categories.

For future work, feature engineering techniques could be explored to better differentiate features in both pure and mixed samples of the *Mambo* class for improved pattern recognition. Additionally, exploring attention-based and transformer architectures such as Vision Transformers (ViTs) could potentially enhance performance further. Real-time deployment feasibility, including latency and inference speed under constrained hardware, also warrants future investigation. Lastly, adding more drone models to the dataset and utilizing pre-trained CNN-based architectures such as VGG-19 may contribute to increased performance in a wider range of drone samples.

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